Module 7, Lesson 1
AI 101 for Mechanical Engineers
Fit a line and interpret coefficients

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Lesson Outline

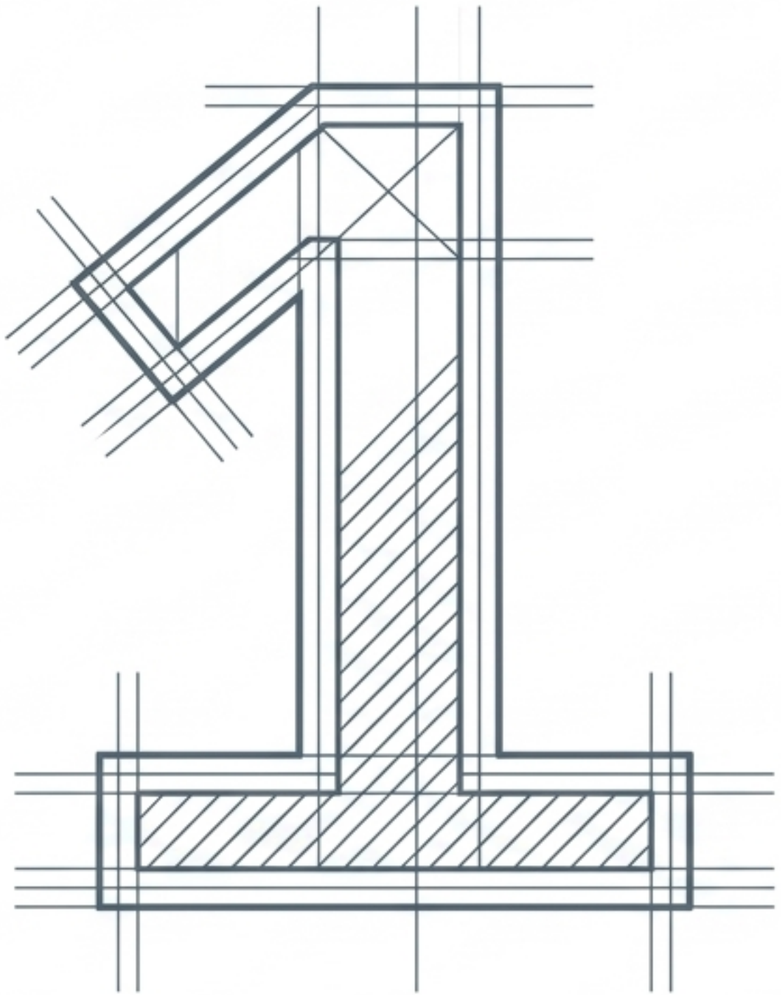
1. Linear Response Models

2. Prediction and Vocabulary

3. Applied Concepts and Reference

4. Knowledge Validation

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1. Linear Response Models

Define the linear regression model, physical variables, and the least-squares fitting criterion.

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The Linear Response Equation

ΔT: Temperature rise in Kelvin (K).

P: Heater power in Watts (W).

$$\Delta T = b_0 + b_1 P$$

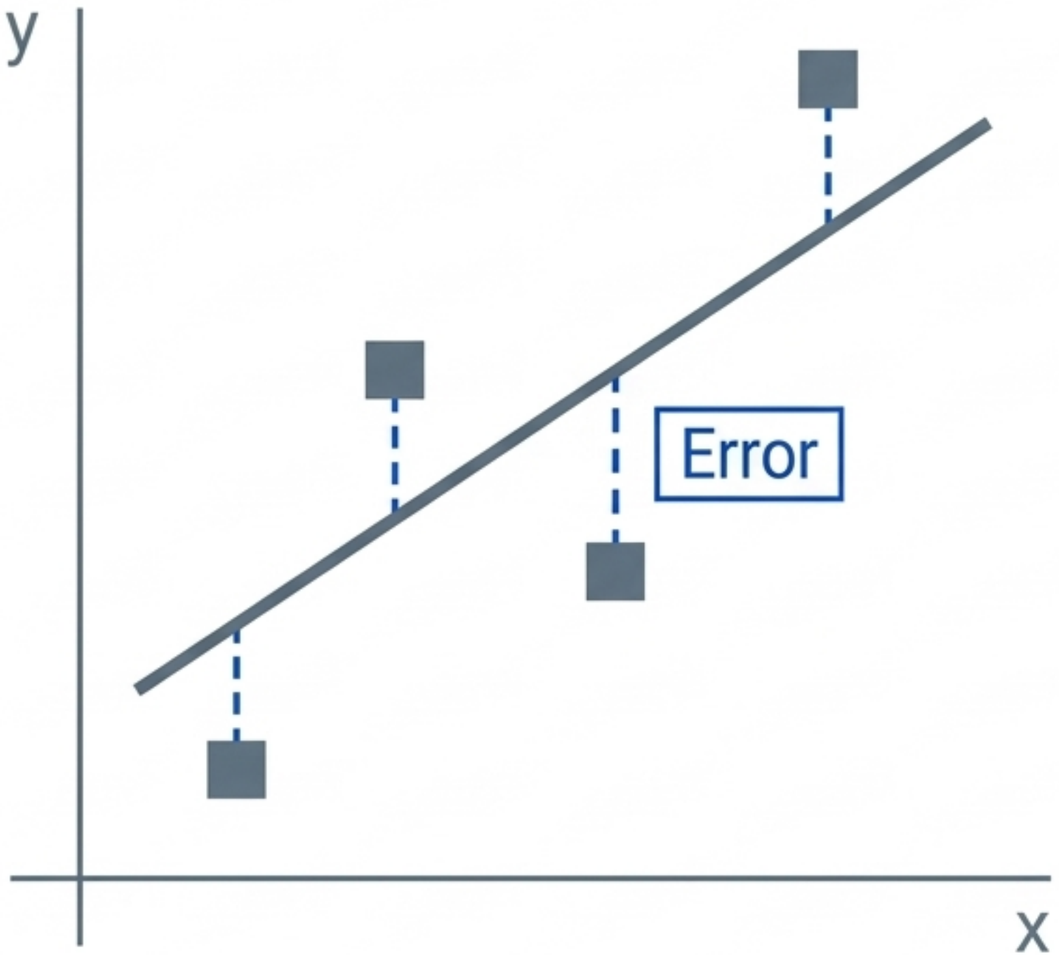
b_1: Slope coefficient in Kelvin per Watt.

b_0: Intercept constant in Kelvin (K).

Physical interpretation requires explicit measurement design assumptions.

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Least Squares and Linearity



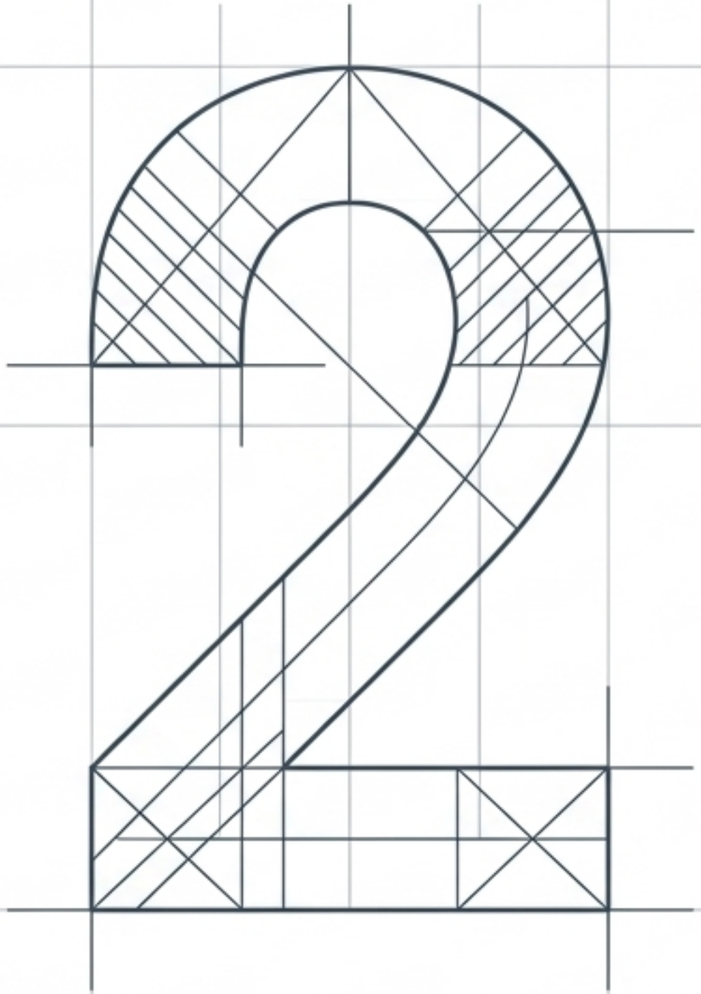
Minimizes sum of squared observation-minus-prediction errors.

Establishes mathematical linearity within fitted coefficients.

Permits polynomial fitting while maintaining coefficient linearity.

Straight-line models ensure visible predictions and errors.

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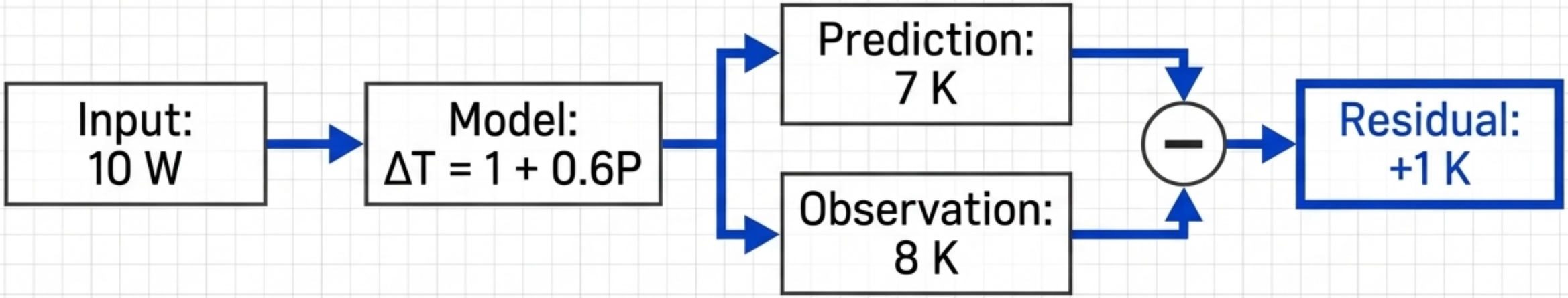


2. Prediction and Vocabulary

Apply the linear model to predict system responses and define foundational regression terminology.

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Calculating Predictions and Residuals



Fitted model equation evaluates: $\Delta T = 1 + 0.6P$.

Inputting 10 W yields a 7 K predicted rise.

Measured rise of 8 K yields a +1 K residual.

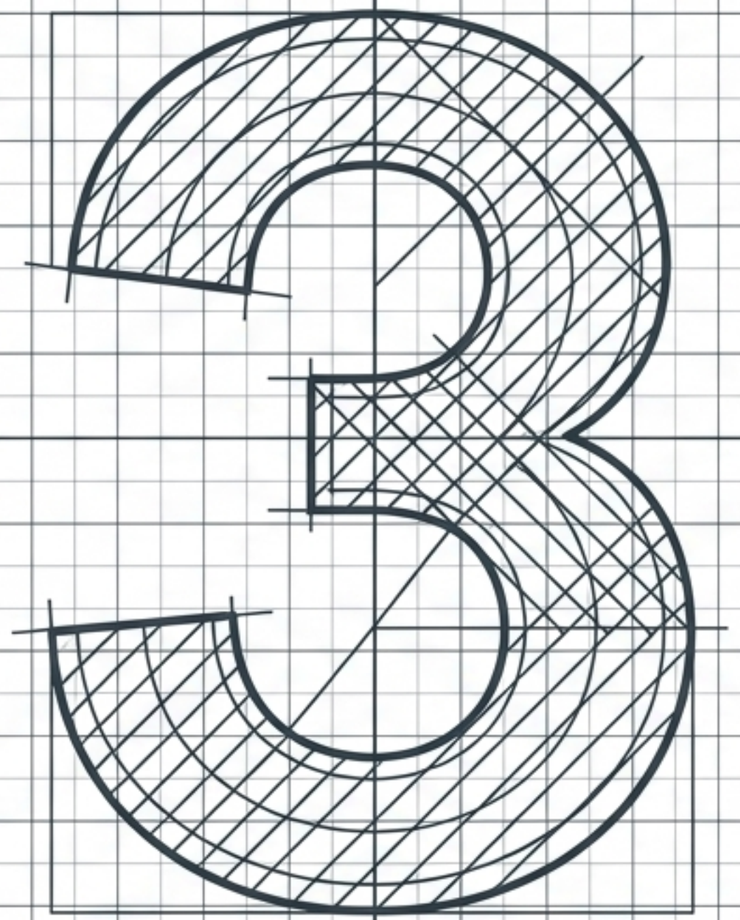
Parameter adjustments alter hypotheses, not experimental data.

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Formal Regression Terminology

Linear regression	Model mathematically linear in its fitted coefficients.
Coefficient	Fitted numerical factor multiplying a specific model term.
Intercept	Additive constant indicating predicted output at zero input.
Residual	Observation value minus the model prediction value.

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3. Applied Concepts and Reference

Determine rigorous SI unit requirements for coefficients and consolidate physical reference parameters.

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Dimensional Analysis of Coefficients

$$[K] = [K] + \left[\frac{K}{W}\right] \times [W]$$

System output explicitly requires Kelvin (K).

Input parameter provides power in Watts (W).

Intercept constant must have units of Kelvin (K).

Slope must have units of Kelvin per Watt.

Multiplication cancels Watts, yielding required Kelvin output.

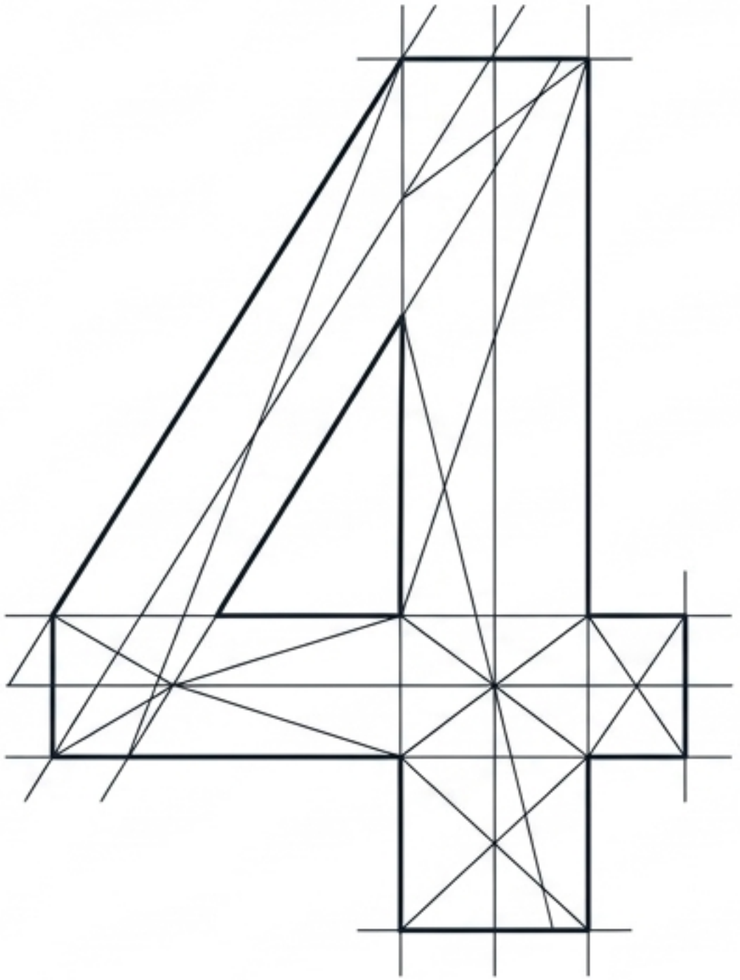
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Consolidated Concept Reference

Model: $\Delta T = 1 + 0.6P$

System Parameter	Calculated Value
Model Intercept	1 K
Model Slope	0.6 K/W
Predicted Rise at 10 W	7 K
Measured Observation	8 K
Calculated Residual	+1 K

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4. Knowledge Validation

Assess comprehension of numerical predictions, residual error calculations, and output-to-input unit derivations.

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Prediction and Residual Validation

Check 1: Predict temperature rise at 10 W input.

Result 1: 7 K (1.6 × 10 = 7).

Check 2: Calculate residual for 8 K measured output.

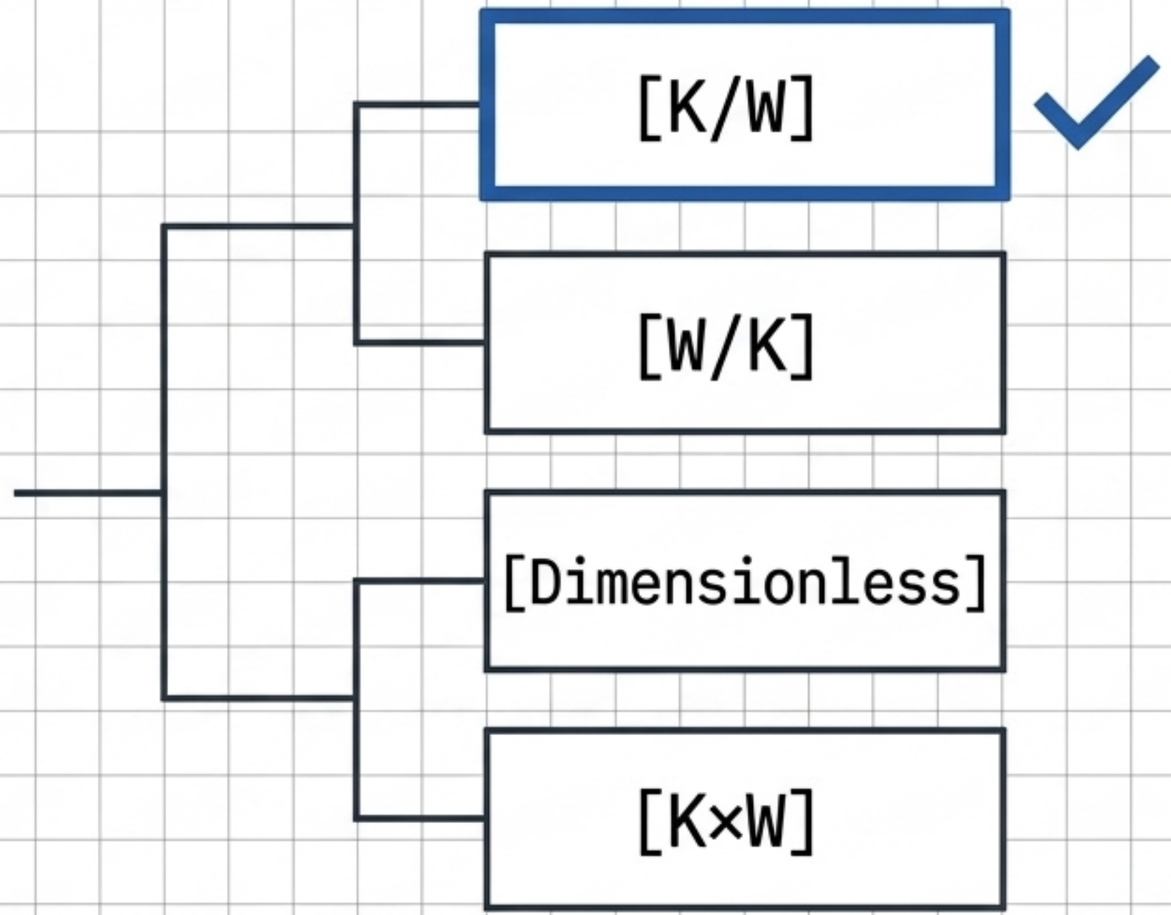
Result 2: +1 K (indicating model underprediction error).

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Slope Unit Validation

Check 3: Identify slope units predicting K from W.
Provided options: K/W, W/K, Dimensionless, or K×W.

Result 3: K/W is the correct analytical unit.
Represents output unit change per input unit change.



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Key Points

[1] Least squares minimizes squared residuals for linear models.

[2] Regression parameters enable precise predictions and residual calculations.

[3] Mathematical consistency requires strict adherence to SI units.

[4] Validation requires calculating numerical predictions and residual errors.

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